

UNDERSTANDING REASONING IN THINKING LANGUAGE MODELS VIA STEERING VECTORS

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ABSTRACT

Recent advances in large language models (LLMs) have led to the development of *thinking* language models that generate extensive internal reasoning chains before producing responses. While these models achieve improved performance, controlling their reasoning processes remains challenging. This work presents a steering approach for thinking LLMs by analyzing and manipulating specific reasoning behaviors in DeepSeek-R1-Distill models. Through a systematic experiment on 500 tasks across 10 diverse categories, we identify several reasoning behaviors exhibited by thinking models, including expressing uncertainty, generating examples for hypothesis validation, and backtracking in reasoning chains. We demonstrate that these behaviors are mediated by linear directions in the model’s activation space and can be controlled using steering vectors. By extracting and applying these vectors, we provide a method to modulate specific aspects of the model’s reasoning process, such as its tendency to backtrack or express uncertainty. Our approach offers practical tools for steering reasoning processes in thinking models in a controlled and interpretable manner. We validate our steering method using three DeepSeek-R1-Distill models, demonstrating consistent control across different model architectures.

1 INTRODUCTION

A recent trend in the development of large language models (LLMs) has been *thinking* LLMs, which generate extensive internal reasoning chains before producing responses (Reynolds & McDonell, 2021; Nye et al., 2021; Wei et al., 2022). Examples include OpenAI’s o1 (OpenAI, 2024) and DeepSeek’s R1 (DeepSeek-AI, 2025). These models have achieved remarkable improvements in performance (Chollet, 2024), yet controlling and understanding their internal reasoning processes remains challenging.

To address this challenge, we develop a steering approach for thinking LLMs by analyzing reasoning behaviors in DeepSeek-R1-Distill models. We focus on several reasoning behaviors exhibited by these models, including their tendency to express uncertainty, backtrack in reasoning chains, and generate examples for hypothesis testing. While these behaviors may not constitute a complete taxonomy of reasoning mechanisms, they provide a practical foundation for developing steering methods.

We investigate whether these reasoning behaviors of thinking LLMs can be directly controlled using *steering vectors*, a method that has been shown to allow precise behavioral control in LLMs (Subramani et al., 2022; Turner et al., 2023; Zou et al., 2023; Panickssery et al., 2023; Templeton et al., 2024; Ardit et al., 2024a). By extracting and applying steering vectors, we provide a means to modulate the internal reasoning dynamics of thinking LLMs in a controlled manner.

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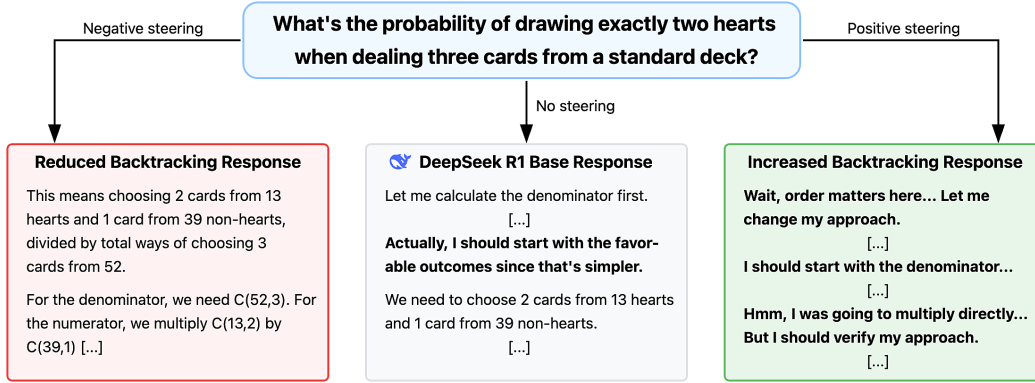


Figure 1: *Steering* on DeepSeek-R1’s *backtracking* feature vector changes the model’s behavior. Depending on whether we add or subtract this vector to the activations at inference time, the model increases or decreases its tendency to abandon its current approach and explore alternative strategies for the task at hand. Highlighted sections indicate instances of this behavior.

In summary, our work presents the following key contributions:

1. We develop a method to extract steering vectors for specific reasoning behaviors in DeepSeek-R1-Distill models, enabling precise control over aspects of their thinking process, such as *backtracking* and *uncertainty estimation*.
2. We demonstrate the effectiveness of our steering approach through an empirical evaluation on over 500 tasks across 10 diverse categories, showing consistent control across multiple DeepSeek-R1-Distill model architectures and sizes.

Our steering approach provides a practical method for controlling reasoning processes in thinking models in an interpretable manner, opening new possibilities for fine-grained control of model behavior. To facilitate reproducibility and enable further research, we make our complete experimental codebase and datasets publicly available.¹

2 PRELIMINARIES

2.1 ATTRIBUTION PATCHING

A fundamental challenge in analyzing the behavior of large language models (LLMs) is identifying the specific components and layers responsible for a given behavior. A widely used technique for addressing this challenge is *activation patching* (Meng et al., 2022). Activation patching works by replacing the activations of a specific model component with those from a *counterfactual* example, which differs only in a specific aspect of the behavior being analyzed. If this intervention significantly alters the model’s output with respect to the observed behavior, the modified component can be marked as playing a key role in implementing this behavior. The *patching effect* is quantified as the change in a relevant output metric:

$$\Delta L = L(\mathbf{x}_{\text{clean}} \mid \text{do}(\mathbf{a} = \mathbf{a}_{\text{patch}})) - L(\mathbf{x}_{\text{clean}}),$$

where L is a metric measuring the difference in model outputs (e.g., KL-divergence), $\mathbf{a}$ is the original activation, and $\mathbf{a}_{\text{patch}}$ is the counterfactual activation.

Since activation patching is computationally expensive, a more efficient linear approximation known as *attribution patching* (Nanda, 2023; Syed et al., 2023) is often used, which utilizes the gradients of the model’s activations with respect to the metric:

$$\Delta L \approx (\mathbf{a}_{\text{patch}} - \mathbf{a}_{\text{clean}})^T \cdot \frac{\partial}{\partial \mathbf{a}_{\text{clean}}} L(\mathbf{x}_{\text{clean}} \mid \text{do}(\mathbf{a} = \mathbf{a}_{\text{clean}})).$$

¹<https://github.com/cvenhoff/steering-thinking-llms>

2.2 COMPUTING STEERING VECTORS

The *Difference of Means* method is a widely used technique for extracting steering vectors in LLMs (Turner et al., 2024; Arditi et al., 2024b). This technique is based on constructing contrastive datasets that differ in a specific concept and computing the difference in their mean activations of a model. Formally, let D_+ and D_- be two datasets where samples in D_+ exhibit a given concept, while samples in D_- do not. Given a model component, we compute the *Difference of Means* vector as:

$$\mathbf{u} = \frac{1}{|D_+|} \sum_{p_i \in D_+} \mathbf{a}(p_i) - \frac{1}{|D_-|} \sum_{p_j \in D_-} \mathbf{a}(p_j)$$

where $a(p_i)$ and $a(p_j)$ represent the activations of the model components over the prompts from the respective datasets. This vector u captures the primary direction in activation space that differentiates the two datasets with respect to the target concept. In cases where explicitly matched counterfactuals are unavailable, a common heuristic is to define D_+ as the set of all samples exhibiting the target behavior, while D_- consists of the full dataset. In this scenario, the *Difference of Means* vector is computed by subtracting the overall mean activation from the mean activation of the behavior-associated examples. This isolates the direction in the activation space most associated with the target behavior while reducing the influence of general model biases.

2.3 DEEPSEEK THINKING MODELS

As mentioned in the introduction, *thinking* models are a type of language model designed to generate long chains of internal reasoning before arriving at a final answer. Examples of this type of models include QwQ (Qwen Team, 2024), Gemini 2.0 Flash Thinking (GDM, 2024), o1 (OpenAI, 2024), and DeepSeek-R1 (DeepSeek-AI, 2025).

In this work, we focus on characterizing the thinking mechanisms of DeepSeek-R1, a recent *thinking* model that has achieved a similar performance to o1-preview on the ARC-AGI-Pub dataset (Knoop, 2025; Chollet, 2024). DeepSeek-R1 is a language model trained through a multi-stage process that combines large-scale reinforcement learning (RL) with the strategic use of supervised fine-tuning (SFT). The model’s architecture uses a Mixture-of-Experts (MoE) approach with 37B activated parameters and 671B total parameters.

The DeepSeek team has distilled R1’s reasoning capabilities into smaller, dense models ranging from 1.5B to 70B parameters, based on both Qwen and Llama architectures. These distilled models achieve similar or better performance than frontier production models like GPT-4o and Claude 3.5 Sonnet at several math and coding benchmarks (DeepSeek-AI, 2025). We use the Qwen-14B and Llama-8B distilled models of DeepSeek R1 for our analysis.

3 IDENTIFYING REASONING BEHAVIORS FOR STEERING

To identify reasoning behaviors suitable for steering, we examined 100 reasoning chains generated by DeepSeek-R1 and 100 answers produced by GPT-4o, using tasks across diverse categories (see Section 4.1). Consistent with prior findings (DeepSeek-AI, 2025), we observe that one of the key differences is that the DeepSeek-R1 distills tend to explore many different approaches when solving a task, whereas the base models follow a more linear reasoning trajectory. Notably, the thinking models can express uncertainty about its current approach, generate examples or scenarios to test hypotheses, and backtrack to intermediate steps when revising its reasoning.

Based on these observations, we identify the following behavioral patterns in the DeepSeek-R1 distill reasoning processes as targets for our steering approach:

- **Initialization:** The model rephrases the task and articulates initial thoughts, typically at the beginning of the reasoning chain.
- **Deduction:** The model derives conclusions based on its current approach and assumptions.
- **Knowledge Augmentation:** The model incorporates external knowledge to refine its reasoning.
- **Example Testing:** The model generates examples or scenarios to validate its working hypothesis.